Supervision and reference

Stop/work supervisor
drive to station / mow with arm

Cutter reference trajectory
truck frame, head down, 0.16–0.18 m/s

line / arc / circle / sine segments,
cosine cruise and turnaround profiles

Task-space control layer

Task-space error + EE cascade
cylinders 1–3 position, cylinder 4 holds head down

+

Residual PPO policy
bounded correction,
cylinders 1–3 only

Cylinder-space DLS
damped inverse of measured 3×4 cutter Jacobian

Desired cylinder command c = [Δx*, Δxdot*] (8-D)

measured state x, xdot, q, p_ → policy observations

Learned actuator realization — the experimental variable between PPO1 and PPO2

PPO1 actuator path
paired ID–FD hybrid:
0.8 requested increment + 0.2 learned residual
FD(ID(target)) − FD(ID(zero target))

one path
active per
experiment

PPO2 actuator path
command-conditioned hydraulic ROM:
internal valve → Δp → xdot → x
GRU encoder + physical update laws

Bounded PhysX cylinder target

Simulated plant and feedback

PhysX articulation of the FME 500
IsaacLab, 1/120 s physics, 60 Hz control

reference / policy command
task-layer signal flow
learned actuator realization
plant actuation
measured feedback (60 Hz)